

Experimental Report 03

Safety-Oriented RUL Evaluation and Robust Calibration

Remaining Useful Life prediction from multivariate sensor time series

Evaluation focus
safety-oriented RUL

Best calibrated MAE
9.48 cycles

Calibration gain
13.38 → 12.50 MAE

Target cap range
120–130 cycles

Safety-oriented experiment setup

- The experimental focus moved from pure regression accuracy toward safety-oriented RUL evaluation
- Overestimation was treated as the most critical error type: predicting failure later can create operational risk
- Evaluation included MAE, overestimation behavior, near-failure behavior, conservative calibration, and RUL target shaping
- Capped RUL targets reduce noise at high remaining-life values; the cap was useful but sensitive to tuning

RUL zone definitions

Zone	Definition	Interpretation
Near-failure	true RUL ≤ 30	highest operational risk
Warning zone	30 < true RUL ≤ 70	maintenance planning range
Mid-life	70 < true RUL ≤ 120	intermediate degradation
Long RUL	true RUL > 120	noisier high-RUL region

Core experiment results

Experiment direction	Main idea	Observed result
CatBoost comparison	Test whether CatBoost can better use aggregate and lag-based features.	Did not improve RF; CatBoost MAE ≈13.5 vs raw-window RF ≈13.2.
Safety analysis	Treat overestimation as a safety-critical error.	Evaluation expanded beyond MAE.
Conservative calibration	Shift predictions downward to reduce unsafe overestimation.	Aggregated RF MAE improved 13.38 → 12.50.
Validation-only evolution	Select one strongest validation winner.	Exposed a failure case on official test.
Robust evolutionary search	Compare robust candidates instead of one validation winner.	Best official calibrated MAE reached 9.48.
Baseline / health features	Compare sensor values with each engine’s early state.	Improved interpretability and reduced overestimation.

MAE evolution across main approaches

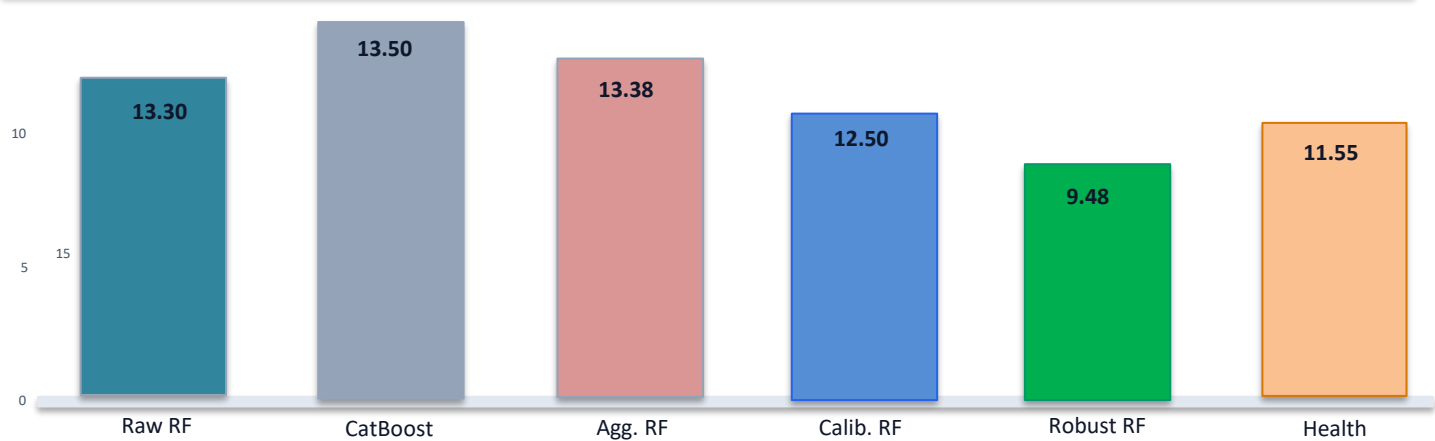


Figure 1. Main reported MAE stages. Values above bars are MAE; labels below bars show model directions.

Conservative Calibration and Robust Search

The central improvement was methodological: calibration and robust candidate selection replaced one-off validation winners.

Conservative calibration

calibrated_prediction = max (original_prediction – shift, 0)

Model	Before calibration	Shift	After calibration	Overestimation share
Raw-window RF	13.30	4.5	12.88	0.49
Aggregated-feature RF	13.38	4.6	12.50	0.50

Capped RUL targets were useful but sensitive to tuning. The most stable tested cap values were around 120–130 cycles, especially 125

Robust evolutionary feature search

- The first evolutionary search selected one best validation candidate, but this did not transfer reliably to the official test.
- The improved approach used evolutionary search as a generator of promising configurations.
- Candidates were compared across validation fitness, validation MAE, near-failure MAE, low overestimation, different RUL caps, and different window sizes.

Best robust RF configuration

Parameter	Value
Window size	40
RUL cap	125
Feature count	384
Conservative shift	2
Official calibrated MAE	9.48
RMSE	13.20
R²	0.899
Overestimation share	0.47
Maximum overestimation	35.48

Best tested direction: robust evolutionary aggregated Random Forest with conservative calibration. Official calibrated MAE improved from 12.50 to 9.48 cycles.

Accuracy-safety trade-off

Model direction	MAE	Overestimation share	Interpretation
Robust RF best MAE	9.48	0.47	best accuracy
Health robust best MAE	11.55	0.37	more conservative
Health robust best safety	12.31	0.32	lowest overestimation

Domain-Aware Sensor Interpretation

Sensor behavior was interpreted with respect to the FD001 setting and engine-control effects.

Observed degradation behavior

- FD001 uses fixed operating conditions, so several sensors can remain nearly constant and carry little degradation information
- Engine degradation is not a smooth monotonic decline: sensor traces can move in zigzags and may temporarily look healthier
- The first tens of cycles may look close to normal because early-stage wear is weak or not visible in sensor readings
- An active control system can keep some sensor values near nominal conditions until later degradation stages
- This supports filtering, windowed features, baseline normalization, and safety-aware calibration rather than raw sensor trends alone

Sensor relevance in the FD001 setting

Group	Sensors	Physical meaning / interpretation
More informative	sensor_4 / column 9 sensor_9 / column 14 sensor_11 / column 16 sensor_14 / column 19	Low-pressure turbine outlet temperature; physical core speed; static pressure at high-pressure compressor outlet; corrected core speed. More relevant to compressor degradation and engine health changes.
Less informative in this dataset	sensor_1 / column 6 sensor_5 / column 10 sensor_10 / column 15 sensor_16 / column 21 sensor_18 / column 23	In fixed-condition FD001, some signals may be constant or actively controlled. They are not universally useless, but less informative for this specific setting.

Safety-oriented modeling logic

Design choice	Reason for inclusion
Overestimation analysis	Predicting too much remaining life is more dangerous because delayed maintenance can increase failure risk.
Near-failure evaluation	Errors close to failure are operationally more important than errors far from failure.
Conservative calibration	A downward shift reduces unsafe overestimation while preserving useful ranking behavior.
Capped RUL target	High-RUL values can be noisy; a cap around 120–130 cycles improved stability.
Robust feature search	Validation-only winners can fail to transfer; robust search prioritizes reliability under stronger checks.

Report summary

Aspect	Result
Evaluation shift	from pure MAE optimization to safety-oriented RUL evaluation
Calibration result	aggregated RandomForest MAE improved from 13.38 to 12.50
Best result	robust evolutionary aggregated RandomForest reached official calibrated MAE 9.48
Alternative feature direction	baseline / delta / health features improved interpretability and reduced overestimation
Sensor insight	fixed operating conditions and active control make some sensors weak or misleading until late degradation